

Vehicle Safety Systems

Vehicle safety systems are engineering features that reduce crash risk and reduce injury when a crash occurs. Modern vehicles combine mechanical design, sensors, electronic control units, braking systems, warning interfaces, and protective structures.

Active safety systems help prevent an accident before impact. Anti-lock braking systems reduce wheel lockup during hard braking. Electronic stability control helps a driver maintain control by adjusting braking or engine power when the vehicle begins to skid. Traction control limits wheel spin during acceleration.

Advanced driver assistance systems are also active safety features. Lane departure warning alerts the driver when the vehicle drifts from its lane. Blind spot monitoring warns about vehicles beside or behind the car. Forward collision warning and automatic emergency braking can detect a potential collision and apply braking when the driver does not react quickly.

Passive safety systems protect occupants after a collision begins. Seat belts restrain occupants and reduce the chance of ejection. Airbags cushion impact forces. Crumple zones absorb crash energy by deforming in controlled ways. Strong passenger compartments help preserve survival space.

A complete safety strategy uses active and passive systems together. Active systems reduce the likelihood of a crash, while passive systems reduce harm if the crash cannot be avoided. Safety performance also depends on driver behavior, road conditions, maintenance, and regulatory testing.